

Problem 15

Verify the correctness of the following equalities:

$$\begin{aligned} 1^\circ \overline{\overline{AB}} &= A + B; & 2^\circ \overline{\overline{A} + \overline{B}} &= AB; & 3^\circ A + B &= AB + A\Delta B; \\ 4^\circ \overline{A\Delta B} &= AB + \overline{AB}; & 5^\circ A\Delta B &= (\overline{AB})\Delta(\overline{AB}); \\ 6^\circ \overline{\bigcup_{i=1}^n A_i} &= \bigcap_{i=1}^n \overline{A_i}; & 7^\circ \overline{\bigcap_{i=1}^n A_i} &= \bigcup_{i=1}^n \overline{A_i}. \end{aligned}$$

Note: $A\Delta B$ denotes the symmetric difference: $A\Delta B = (A\overline{B}) + (\overline{A}B)$

Solution

Part 1°: Verify $\overline{\overline{AB}} = A + B$

Proof

Using De Morgan's law: $\overline{XY} = \overline{X} + \overline{Y}$

$$\overline{\overline{AB}} = \overline{\overline{A} + \overline{B}} = A + B$$

TRUE

This is De Morgan's law applied to the product $\overline{AB}$.

Part 2°: Verify $\overline{\overline{A} + \overline{B}} = AB$

Proof

Using De Morgan's law: $\overline{X + Y} = \overline{X}\overline{Y}$

$$\overline{\overline{A} + \overline{B}} = \overline{\overline{A}} \cdot \overline{\overline{B}} = A \cdot B = AB$$

TRUE

This is De Morgan's law applied to the sum $\overline{A} + \overline{B}$.

Part 3°: Verify $A + B = AB + A\Delta B$

Proof

Recall: $A\Delta B = A\overline{B} + \overline{A}B$ (symmetric difference)

Right side:

$$AB + A\Delta B = AB + A\overline{B} + \overline{A}B$$

Factor:

$$= A(B + \overline{B}) + \overline{A}B$$

$$= A \cdot 1 + \overline{A}B = A + \overline{A}B$$

Using absorption ($X + \overline{X}Y = X + Y$):

$$= A + B$$

TRUE

Interpretation: $A \cup B$ can be decomposed into the intersection $A \cap B$ plus the symmetric difference (elements in exactly one set).

Part 4°: Verify $\overline{A \triangle B} = AB + \bar{A}\bar{B}$

Proof

Start with the left side:

$$\overline{A \triangle B} = \overline{AB + \bar{A}\bar{B}}$$

Using De Morgan's law:

$$= \overline{AB} \cdot \overline{\bar{A}\bar{B}}$$

$$= (\bar{A} + B)(A + \bar{B})$$

Expand:

$$= \bar{A}A + \bar{A}\bar{B} + BA + B\bar{B}$$

$$= \emptyset + \bar{A}\bar{B} + AB + \emptyset$$

$$= AB + \bar{A}\bar{B}$$

TRUE

Interpretation: The complement of symmetric difference includes cases where both events occur or both don't occur.

Part 5°: Verify $A \triangle B = (\overline{A\bar{B}}) \triangle (\overline{\bar{A}B})$

Proof

First, simplify the right side terms:

$$\overline{A\bar{B}} = \bar{A} + B$$

$$\overline{\bar{A}B} = A + \bar{B}$$

So:

$$(\overline{A\bar{B}}) \triangle (\overline{\bar{A}B}) = (\bar{A} + B) \triangle (A + \bar{B})$$

Using the definition of symmetric difference:

$$= (\bar{A} + B)(\overline{A + \bar{B}}) + \overline{(\bar{A} + B)(A + \bar{B})}$$

$$= (\bar{A} + B)(\bar{A} \cdot B) + (A\bar{B})(A + \bar{B})$$

Wait, let me recalculate more carefully.

$$(\bar{A} + B) \triangle (A + \bar{B}) = [(\bar{A} + B) \cap \overline{(A + \bar{B})}] \cup [\overline{(\bar{A} + B)} \cap (A + \bar{B})]$$

$$= [(\bar{A} + B)(\bar{A}B)] + [(A\bar{B})(A + \bar{B})]$$

$$= [(\bar{A} + B) \cap (\bar{A} \cap B)] + [(A \cap \bar{B}) \cap (A + \bar{B})]$$

This is getting complex. Let me use a direct approach:

$$(\bar{A} + B) \cap \overline{(A + \bar{B})} = (\bar{A} + B)(\bar{A}B)$$

Since $\bar{A}B \subseteq (\bar{A} + B)$:

$$= \bar{A}B$$

Similarly:

$$\overline{(\bar{A} + B)} \cap (A + \bar{B}) = (A\bar{B})(A + \bar{B}) = A\bar{B}$$

Therefore:

$$(\bar{A} + B) \triangle (A + \bar{B}) = \bar{A}B + A\bar{B} = A \triangle B$$

TRUE

Part 6°: Verify $\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \bar{A}_i$

Proof

This is De Morgan's law for unions:

$$\overline{A_1 \cup A_2 \cup \dots \cup A_n} = \bar{A}_1 \cap \bar{A}_2 \cap \dots \cap \bar{A}_n$$

An element is NOT in the union if and only if it's not in ANY of the sets, which means it's in ALL of the complements.

TRUE

This is a fundamental law (De Morgan's law for general unions).

Part 7°: Verify $\overline{\bigcap_{i=1}^n A_i} = \bigcup_{i=1}^n \bar{A}_i$

Proof

This is De Morgan's law for intersections:

$$\overline{A_1 \cap A_2 \cap \dots \cap A_n} = \bar{A}_1 \cup \bar{A}_2 \cup \dots \cup \bar{A}_n$$

An element is NOT in the intersection if and only if it's not in AT LEAST ONE of the sets, which means it's in AT LEAST ONE of the complements.

TRUE

This is De Morgan's law for general intersections.

Summary of Results

Identity	Status	Name/Description
1°	TRUE	De Morgan's law for products
2°	TRUE	De Morgan's law for sums
3°	TRUE	Union decomposition
4°	TRUE	Complement of symmetric difference
5°	TRUE	Symmetric difference identity
6°	TRUE	De Morgan's law (general union)
7°	TRUE	De Morgan's law (general intersection)

Final Answer

ALL IDENTITIES ARE TRUE

Key principles used:

- De Morgan's laws: $\overline{XY} = \bar{X} + \bar{Y}$ and $\overline{X + Y} = \bar{X}\bar{Y}$
- Double complement: $\overline{\bar{A}} = A$

- Absorption: $X + \bar{X}Y = X + Y$
- Symmetric difference: $A \triangle B = A\bar{B} + \bar{A}B$
- Complement laws: $X + \bar{X} = S$ and $X\bar{X} = \emptyset$

All seven identities are standard results in Boolean algebra and set theory.